

YOGESH P,TCE IT
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Project Title and Introduction

Project Title: Secure Blockchain-Based Student Mark Management System

Project Goal: To design and implement a secure, transparent, and immutable system for managing student academic marks using blockchain technology and strong encryption. This project demonstrates proficiency in full-stack development, cryptographic security, and smart contract integration.

Target Audience: School administrators, teachers, and system auditors.

Key Achievement Highlight: Successfully implemented a system where student marks are encrypted on the client side using a unique teacher key derived from Firebase authentication, and then stored immutably on a private Ethereum blockchain.

the secure technology.

Secure Authentication and Key Generation:

Teachers can sign up or log in using Firebase.

A unique symmetric encryption key is derived upon successful login, ensuring only the authenticated teacher can decrypt the marks.

Decentralized Data Management:

Marks are stored in a key-value pair format (e.g., {"Maths": 90}) and then encrypted.

Encrypted records are stored permanently on the blockchain, making them immutable and tamper-proof.

Real-Time Academic Analytics:

The system dynamically decrypts marks upon loading.

Calculates and displays the Total Marks, Class Rank, and Grade for every student in real time.

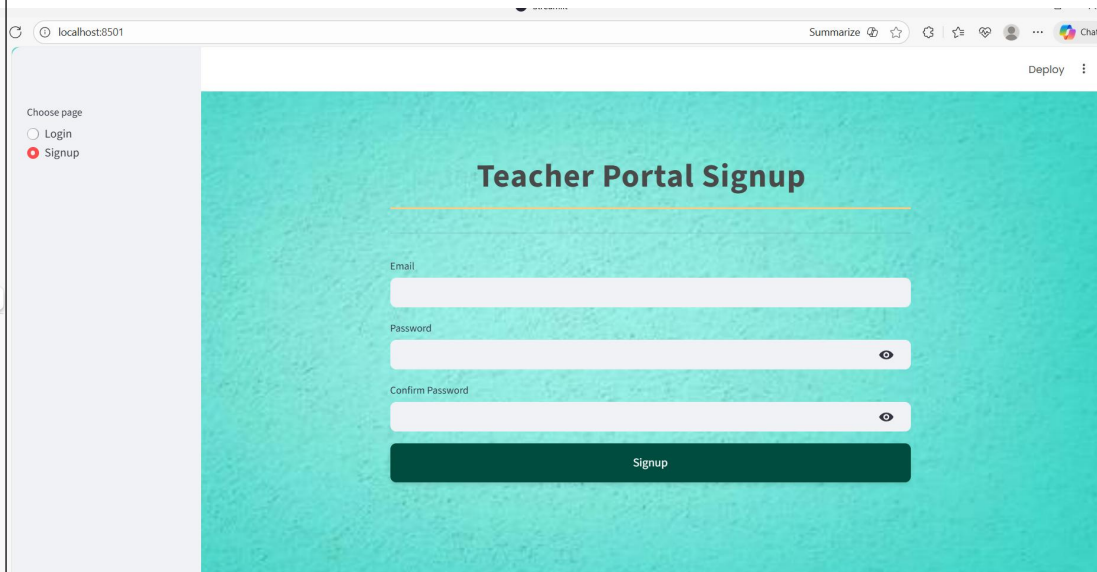
Automatically identifies and highlights the Class Topper (highest marks).

Flexible Data Input with Supports both:

Individual Subject Input: Streamlined forms for creating new students or updating existing student marks subject-by-subject.

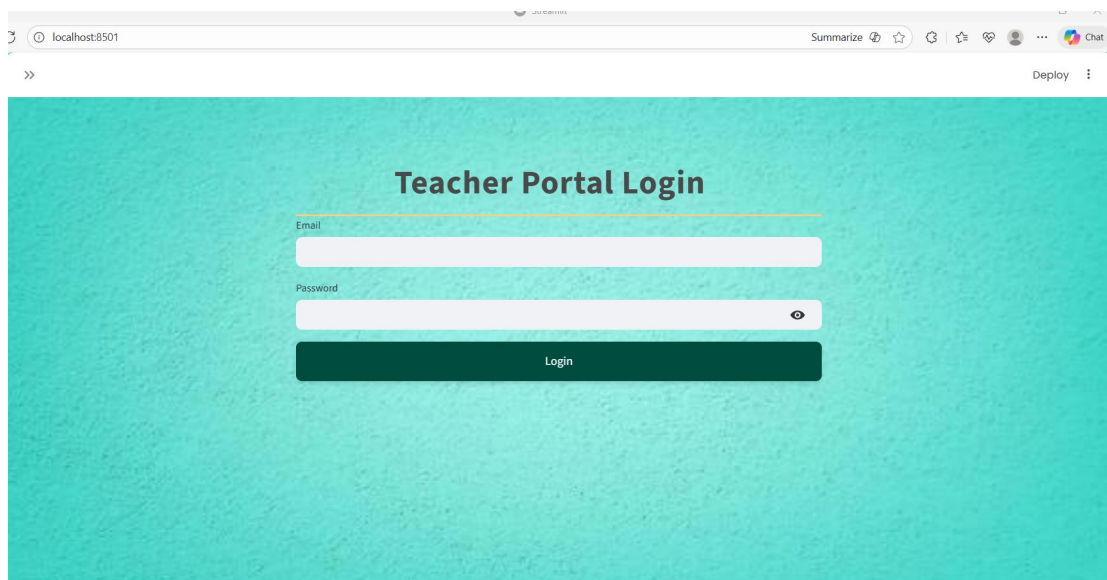
Bulk CSV Upload: Automated batch processing for adding or updating marks for multiple students from a single CSV file.

Signup page:

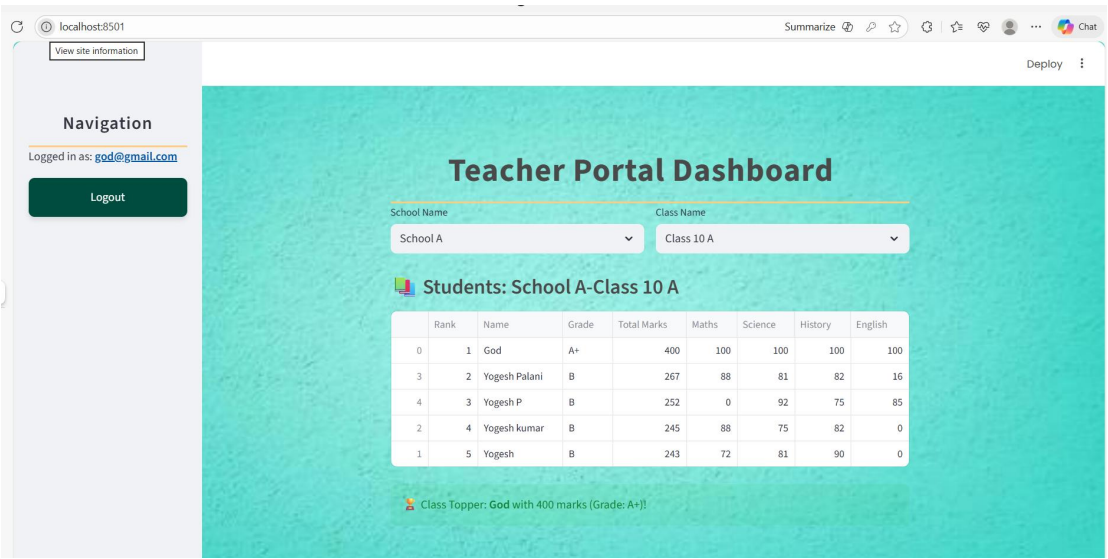
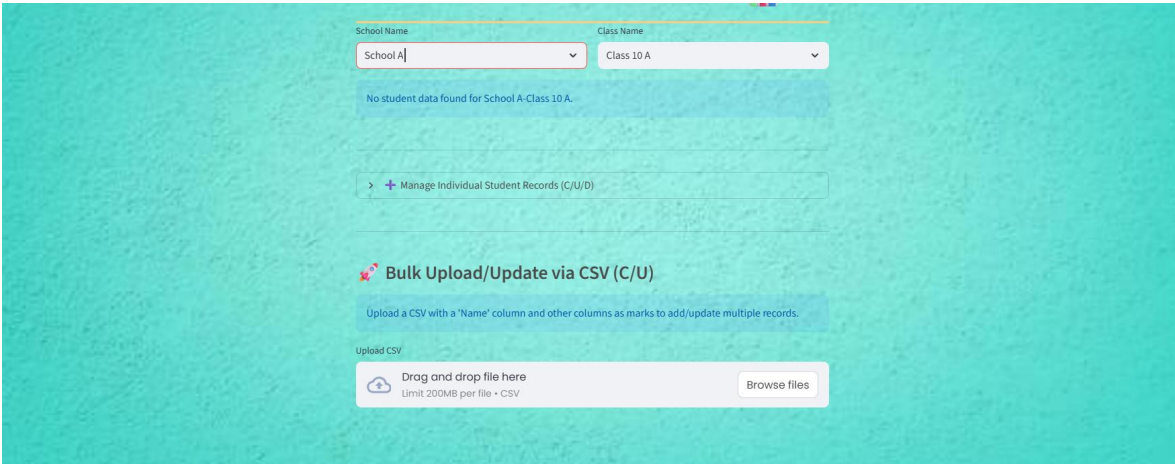


The screenshot shows a web browser window at localhost:8501. On the left, a sidebar contains a 'Choose page' section with radio buttons for 'Login' and 'Signup', where 'Signup' is selected. The main content area has a teal background and features the title 'Teacher Portal Signup' in bold black text. Below the title are three input fields: 'Email', 'Password', and 'Confirm Password'. Each field has a small eye icon to its right. At the bottom of the form is a dark green button labeled 'Signup'.

Login Page:



The screenshot shows a web browser window at localhost:8501. The main content area has a teal background and features the title 'Teacher Portal Login' in bold black text. Below the title are two input fields: 'Email' and 'Password'. Each field has a small eye icon to its right. At the bottom of the form is a dark green button labeled 'Login'.



Update (Modify Existing Marks)

Action: Teacher selects an existing student (e.g., "Alice") and navigates to the "Update Existing Marks" section.

Input Screenshot:

The screenshot shows a web interface for managing student records. At the top, there's a dropdown menu labeled 'Manage Individual Student Records (C/U/D)'. Below it, under 'Select Operation', three radio buttons are visible: 'Add New Student (Create)', 'Update Existing Marks' (which is selected), and 'Delete Student'. The main section is titled 'Update Student Marks'. It features a dropdown menu for 'Select student to update' with 'Yogesh Palani' selected. Below this, a section titled 'Current/New Marks for Yogesh Palani (Max 100)' contains four input fields for 'Maths' (88), 'Science' (81), 'History' (82), and 'English' (16). Each field has minus and plus buttons for adjustment. At the bottom of this section is a button labeled 'Update Marks for Yogesh Palani'.

Create (Add New Student)

Action: Teacher navigates to the "Add New Student (Create)" section.

Input: Enters Student Name and subject marks (e.g., Maths: 95, Science: 88).

Output Screenshot:

The screenshot shows the 'Add New Student' section of the web interface. It starts with the same 'Manage Individual Student Records (C/U/D)' dropdown. Under 'Select Operation', the 'Add New Student (Create)' radio button is selected. The section is titled 'Add New Student'. It includes a text input field for 'New Student Name' with 'ganesh' entered. Below that, a section titled 'Enter Marks (Max 100)' has four input fields for 'Maths' (50), 'Science' (50), 'History' (50), and 'English' (50), each with minus and plus buttons. An 'Add Student' button is located at the bottom of this section.

School Name

Class Name

School A

Class 10 A

Students: School A-Class 10 A

	Rank	Name	Grade	Total Marks	Maths	Science	History	English
0	1	God	A+	400	100	100	100	100
3	2	Yogesh Palani	B	267	88	81	82	16
4	3	Yogesh P	B	252	0	92	75	85
2	4	Yogesh kumar	B	245	88	75	82	0
1	5	Yogesh	B	243	72	81	90	0
5	6	ganesh	C	200	50	50	50	50

🏆 Class Topper: God with 400 marks (Grade: A+)!

Delete (Remove Student Record)

Action: Teacher selects a student (e.g., "Bob") for deletion.

Input Screenshot:

2

4

Yogesh kumar

B

245

88

75

82

0

1

5

Yogesh

B

243

72

81

90

0

5

6

ganesh

C

200

50

50

50

50

🏆 Class Topper: God with 400 marks (Grade: A+)!

+

Manage Individual Student Records (C/U/D)

Select Operation

☐ Add New Student (Create)
 ☐ Update Existing Marks
 ☒ Delete Student

Delete Student Record

Select student to delete

ganesh

Are you sure you want to delete the record for ? This action is permanent on the blockchain.

Confirm Delete:

Read (View Dashboard Data)

Action: Teacher selects the School and Class.

Input: System fetches, decrypts, and processes blockchain data.

Output Screenshot:

School Name
Class Name

School A
Class 10 A

Students: School A-Class 10 A

	Rank	Name	Grade	Total Marks	Maths	Science	History	English
0	1	God	A+	400	100	100	100	100
3	2	Yogesh Palani	B	267	88	81	82	16
4	3	Yogesh P	B	252	0	92	75	85
2	4	Yogesh kumar	B	245	88	75	82	0
1	5	Yogesh	B	243	72	81	90	0

🏆 Class Topper: **God** with 400 marks (Grade: A+)!

> + Manage Individual Student Records (C/U/D)

```

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
✅ View call success: getStudentsBySchoolClass
2025-11-29 21:04:15.101 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
✅ View call success: getStudentsBySchoolClass
✅ Transaction success: 683743961d91a7da8c69f0d91cfa89d4d844a97e042dd357c476f4a6996a51ba
✅ View call success: getStudentsBySchoolClass
2025-11-29 21:04:16.104 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
✅ View call success: getStudentsBySchoolClass
2025-11-29 21:05:47.009 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
✅ View call success: getStudentsBySchoolClass
✅ Transaction success: 1f19a455c62eb70ceffe28384f7d621f83ac0a6645ddaec42434af5515ae62f2
✅ View call success: getStudentsBySchoolClass
2025-11-29 21:05:49.485 Please replace `use_container_width` with `width`.

```

This sequence proves that the application can commit encrypted data to the decentralized ledger, and immediately verify the update by reading the new state of the contract. The hash confirms the data is now permanently recorded and timestamped on the blockchain, fulfilling the project's immutability requirement.

Firebase Authentication Verification

Project Overview

App Check

App Hosting

Authentication

Build

Run

Analytics

AI

Related development tools

Firebase Studio

Spark

Upgrade

Blockchain

Authentication

with Identity Platform

Ask Gemini to help integrate and use Authentication

Users

Sign-in method

Templates

Usage

Settings

Extensions

Protect your Authentication resources from abuse, such as billing fraud or phishing

Configure App Check

The following authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number or user UID

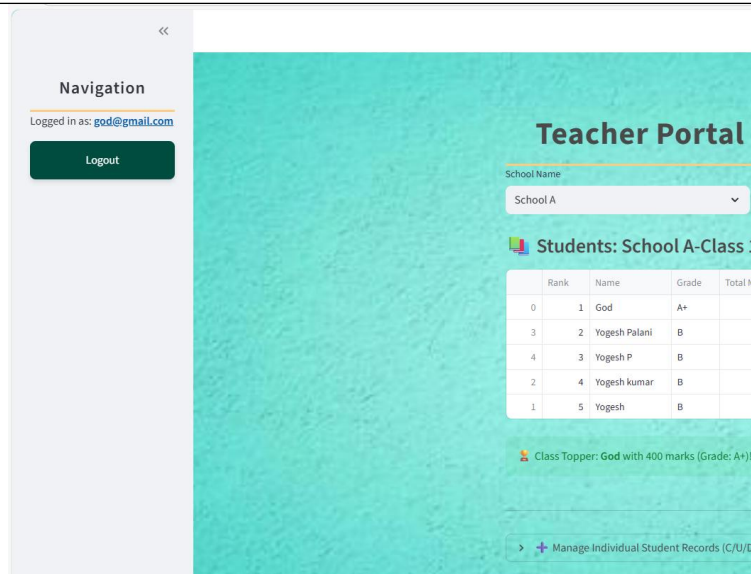
Add user

Identifier	Providers	Created	Signed in	User UID
god@gmail.com		29 Nov 2025		6kj8yilUBJPNLpMUnmRNCs...

Rows per page

50

1 - 1 of 1



Conclusion

The screenshots collectively demonstrate a fully functional, end-to-end decentralized application capable of:

- Securing Access via Firebase.
- Maintaining a Secure Connection to the private blockchain.
- Executing all core CRUD operations (Create, Read, Update, Delete - with Create/Update transactions shown) directly against a Solidity Smart Contract.
- This provides strong evidence of proficiency across Cloud Authentication, Python Application Development, and Blockchain Integration.